$q : [a,b] \to \mathbb{R}$ $\qquad$ $\mathcal{L} : \mathbb{R} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

$q \quad v \quad t$

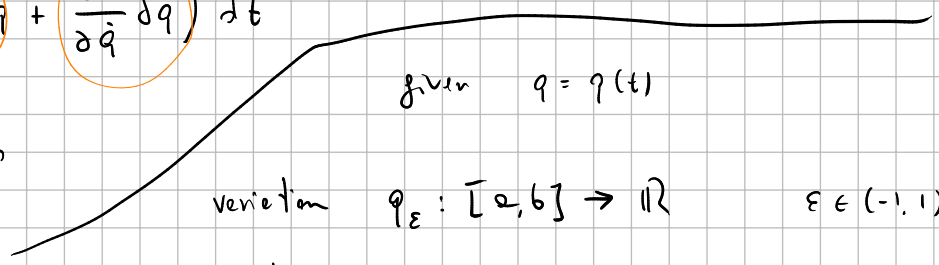
$$S[q] = \int_a^b \mathcal{L}(q(t), \dot{q}(t), t)\, dt$$

physical paths are critical points of S

perturbation of q

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \delta q\right) - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} \delta q$$

$$0 = \delta S = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial \dot{q}} \delta \dot{q}\right) dt$$



given $q = q(t)$

$$\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = 0$$

variation $q_\varepsilon : [a,b] \to \mathbb{R}$ $\qquad \varepsilon \in (-1,1)$

and $q_\varepsilon(a) = q(a) \quad \forall \varepsilon$ $\qquad q_0(t) = q(t) \quad \forall t$

$q_\varepsilon(b) = q(b) \quad \forall \varepsilon$

for all variations

$$0 = \delta S[q] = \frac{d}{d\varepsilon} S[q_\varepsilon]\Big|_{\varepsilon=0}$$

$$= \int_a^b \frac{d}{d\varepsilon} \mathcal{L}(q_\varepsilon(t), \dot{q}_\varepsilon(t), t)\Big|_{\varepsilon=0} dt$$

$$\frac{d}{d\varepsilon} \mathcal{L}(q_\varepsilon(t), \dot{q}_\varepsilon(t), t)\Big|_{\varepsilon=0} = \left[\frac{\partial \mathcal{L}}{\partial q}(q_\varepsilon(t), \dot{q}_\varepsilon(t), t)\, \frac{d q_\varepsilon}{d\varepsilon}(t) + \frac{\partial \mathcal{L}}{\partial v}(q_\varepsilon(t), \dot{q}_\varepsilon(t), t)\, \frac{d}{d\varepsilon}(\dot{q}_\varepsilon)(t)\right]_{\varepsilon=0}$$

$$= \frac{\partial \mathcal{L}}{\partial q}(q(t), \dot{q}(t), t)\, \frac{d q_\varepsilon}{d\varepsilon}\Big|_{\varepsilon=0}(t) + \frac{\partial \mathcal{L}}{\partial v}(q(t), \dot{q}(t), t)\, \frac{d \dot{q}_\varepsilon}{d\varepsilon}\Big|_{\varepsilon=0}(t)$$

$$\delta q(t) = \frac{d}{d\varepsilon} q_\varepsilon\Big|_{\varepsilon=0}(t) \qquad \delta \dot{q}(t) = \frac{d}{d\varepsilon} \dot{q}_\varepsilon\Big|_{\varepsilon=0}(t) = \frac{d}{d\varepsilon}\frac{d}{dt} q_\varepsilon\Big|_{\varepsilon=0}(t) = \frac{d}{dt}\left(\frac{d}{d\varepsilon} q_\varepsilon\Big|_{\varepsilon=0}\right)(t)$$

$$= \left(\frac{d}{dt} \delta q\right)(t)$$

$$\frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial v} \delta \dot{q} = \frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{\partial \mathcal{L}}{\partial v} \frac{d}{dt} \delta q$$

product rule: $\frac{d}{dt}(AB) = \frac{dA}{dt} B + A \frac{dB}{dt} \to A \frac{dB}{dt} = \frac{d}{dt}(AB) - \frac{dA}{dt} B$

$$= \frac{\partial \mathcal{L}}{\partial q} \delta q + \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial v} \delta q\right) - \left(\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial v}\right) \delta q$$

$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial q} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} \right) \delta q \, dt + \underbrace{\int \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \delta q \right) dt}$$

δq arbitrary

$$\frac{\partial \mathcal{L}}{\partial \dot{q}} \delta q \Big]_a^b = 0 \qquad \delta q(b) = \delta q(a) = 0$$

$$q_\varepsilon(a) = q(a)$$

$$\frac{d}{d\varepsilon} q_\varepsilon(a) \Big|_{\varepsilon=0} = 0$$

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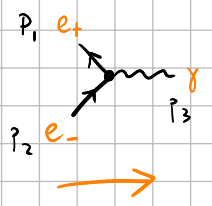
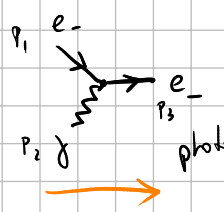
interaction term $\quad ie \bar{\psi} \gamma^\mu A_\mu \psi$ fermion

anti-fermion photon





photoelectric

$$\psi |0\rangle = (\langle 0| \psi^\dagger)^\dagger$$

$$\langle 0| \bar{\psi} A \psi |0\rangle = \langle \bar{\psi} A \psi \rangle = \frac{1}{Z} \int D\bar{\psi} D\psi DA \; e^{iS[\psi, \bar{\psi}, A]}$$

$$\langle 0| \bar{\psi} A \psi |0\rangle = \langle \bar{\psi} A \psi \rangle = \frac{1}{Z} \int D\bar{\psi} D\psi DA \; e^{iS[\psi, \bar{\psi}, A]}$$

$$\langle 0| \bar{\psi} \gamma^\mu A_\mu \psi |0\rangle = \langle 0| A_\mu \bar{\psi} \gamma^\mu \psi |0\rangle$$

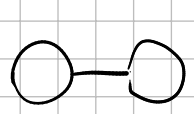
$\rightarrow$ matrices

$[A_\mu, \bar{\psi}_\alpha] = 0$

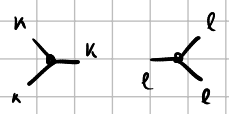
different field commute unless they are both fermions (in which case they anti-commute)

A 4-vector $\qquad A_\mu$ number

γ^μ matrix



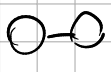
obtained from

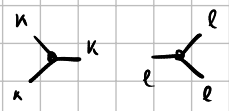


total number $\quad 5!! = 15$

number of partitions into pairs of $2n$ is $(2n-1)!!$

$$\left\langle \sum_k \varphi_k^3 \sum_l \varphi_l^3 \right\rangle_0 = \sum_{k,l} \langle \varphi_k \varphi_k \varphi_k \varphi_l \varphi_l \varphi_l \rangle_0$$

$\sum_{k,l}$ (15 possibilities) $=$ 6  $+$ 9



3!

$$\mathcal{L}(\phi, \partial\phi) = \partial_\mu \phi \partial^\mu \phi - m^2 \phi - V(\phi)$$

$\phi : M \to \mathbb{R}$ $\qquad S[\phi] = \int_M d^4x \, \mathcal{L}(\phi(x), \partial\phi(x))$

canonical quantization: $\phi \Rightarrow \hat{\phi}$ operator

at each $x \in M$ we have $\hat{\phi}(x)$ which acts on Hilbert space

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$[q, p] = i$ $\qquad a = \alpha(q + ip)$ $\qquad a^\dagger = \alpha(q - ip)$

$\Rightarrow [a, a^\dagger] = 1$

ground state $|0\rangle$ $\qquad$ the Hilbert space is built using $a^\dagger$

$|n\rangle \propto (a^\dagger)^n |0\rangle$ $\qquad N = a^\dagger a$ $\qquad N|n\rangle = n|n\rangle$

Hilbert: $\{ |n\rangle \mid n \in \mathbb{N}_0 \}$ $\qquad \langle n | n' \rangle = \delta_{nn'}$

ℓ_2

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scalars $\qquad a(k), \; a^\dagger(k)$ $\qquad$ create / destroy with momentum k

fermions: $\qquad b_s(k), \; b_s(k)^\dagger \to$ c/d with spin s and momentum k are left handed

$d_s(k), \; d_s(k)^\dagger \to$ c/d with spin s and momentum k are right handed

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if $\{b, b^\dagger\} = 1$ $\qquad \{b^\dagger, b^\dagger\} = 0$ $\qquad (b^\dagger)^2 = 0$

$|2\rangle = 0$